SEVEN CANDIDATES / CONTINUATION AUDIT

Recurring signals. Unresolved alternatives.

New local-data and public-archive tests for the seven TESS candidates. This evidence supplement updates the original batch report wherever new results change the assessment. All seven remain unvalidated. The original analysis is retained as provenance.

TIC	Period (d)	Updated assessment
193718989	24.532886	Target favored; nearest deep-EB model constrained by ZTF.
188830138	1.181024	Stronger localization and ZTF constraints; archival RV discrepancy.
270573718	41.383701	Missing FPP completed; substantial nearby-source probability and timing ambiguity.
116046773	5.650272	Real recurrence; target versus 9-arcsec companion unresolved.
382098683	4.351685	Strong host localization; unresolved-companion FPP remains material.
262843927	24.108929	Apparent secondary fails correlated-noise calibration.
201247574	16.560576	Four new later-data events strengthen recurrence.

The changes that matter

TIC 201247574: sectors 102 and 105 contribute four events at the frozen ephemeris, with pooled depth 1.56 +/- 0.30 ppt and S/N 5.13. **TIC 188830138 and TIC 382098683:** host tests now use 20 and 42 sectors. **TIC 270573718:** the missing population calculation now includes nearby and background alternatives; its conditional FPP is materially higher than for most of the sample.

TIC 116046773: more data do not resolve the host; image-template choice reverses the preference. **TIC 262843927:** the phase-0.387 feature has a conditional global p-value of 0.493 despite 8.9 nominal sigma. No robust secondary is established for any candidate.

Units: 1 ppt = 0.001 in relative flux = 0.1%. Probabilities throughout are conditional on the stated data and models. "Planet candidate" is an assessment, not a confirmation or a measured companion mass.

DATA / SCOPE

What was reused, added, and actually tested

The original report, assessment JSON, light-curve and pixel scripts, corrected and aperture time series, Gaia scene audits, PRF calibration, baseline sensitivities and existing FPP caches were read and reused. New outputs live alongside the source analysis in a continuation subfolder, with individual machine-readable results and request receipts.

TIC	Original sectors	Local event windows	Localization sectors	New independent events
193718989	7	8	3	None usable
188830138	19	334	20	None usable
270573718	23	15	4	None usable
116046773	7	25	7	None usable
382098683	42	209	42	None usable
262843927	15	14	3	None usable
201247574	7	10	5	4 (S102/105)

The event counts differ from the original report because this continuation applies its own coverage and 10-minute binning rules. Event windows are admitted by coverage, never by the fitted sign. Retrospective early/late tests share the original discovery alias and event identities; only the later sectors for TIC 201247574 provide genuinely additional TESS recurrence data.

Public archives accessed

MAST/TESSCut; Gaia DR3 source, astrophysical, NSS and available epoch/spectral products; ZTF detection photometry through IRSA; VizieR 2MASS, AllWISE, GALAH, APOGEE and LAMOST; ESO spectral ObsCore; DSS and Pan-STARRS images. Each fetched response has its request parameters, retrieval time, status and SHA-256 hash.

Unavailable or insufficient data are retained as outcomes

ExoFOP endpoints returned a retirement notice or a literal malformed "Array" response. This prevents a complete follow-up census and is not evidence that no observations exist. No measured high-resolution contrast curve or well-sampled precision RV orbit was retrieved. Southern targets lack useful resolved ground transit series in the queried material. Mean Gaia spectra are not epoch RVs.

Gross scene-fit failures are excluded from the main host test at fractional scene RMS > 0.1: TIC 188830138 S14 and TIC 262843927 S31/S88. Their files and failure evidence remain available. TIC 201247574 S103 has no adequately covered primary; that is not a non-detection. Other poor raw-aperture intervals remain visible through degraded search sensitivity.

Recurrence and host localization

TIC	Early predicts late S/N corrected / aperture	Target injection recovery range	Closest alternate conditional tail p range
193718989	4.91 / 2.31	94.20% - 98.60%	0.000999 - 0.015
188830138	8.71 / 8.37	100.00% - 100.00%	0.000999 - 0.000999
270573718	0.29 / 8.19	99.40% - 100.00%	0.000999 - 0.000999
116046773	12.30 / 9.21	34.10% - 95.40%	0.000999 - 1
382098683	7.09 / 5.97	100.00% - 100.00%	0.000999 - 0.000999
262843927	9.09 / 4.11	80.90% - 95.90%	0.000999 - 0.000999
201247574	3.32 / 3.08	79.70% - 93.00%	0.000999 - 0.00999

Local linear baselines are projected out at every transit-model evaluation. Event scatter and a binned-residual inflation factor enter the uncertainties. Multiple starting points reduce timing-fit local-minimum failures. These error models still do not integrate all reduction, stellar, and timing uncertainty.

The corrected early-half ephemeris for TIC 270573718 predicts the later signal poorly (S/N 0.29); the independently extracted aperture series predicts it at S/N 8.19. This discrepancy is reported as timing/reduction sensitivity. It is not silently averaged into a clean ephemeris or labeled a failed astrophysical recurrence.

Image control experiments

For each usable image sector, 128 off-phase decrement maps estimate spatially correlated noise. Catalog-host injections use 1000 realizations per host, two PRF families, two covariance shrinkage settings and five 0.1-pixel astrometric offsets. Catalog neighbors within 65 arcsec are tested when common coverage exists. Per-sector losses and leave-one-sector-out diagnostic sums are saved.

The conditional tail p asks whether an injected alternate host can produce at least the observed preference for the target. Its floor is 1/1001. It is neither a source posterior nor an FPP. Overlapping controls, assumed nominal amplitude, incomplete PRF families, missing Gaia sources and unresolved companions limit any exclusion.

TIC 116046773: the closest companion has loss differences spanning negative and positive values. Cross-template injections recover the target with only 34-95% efficiency, depending on assumptions. This host ambiguity survives the larger seven-sector data set.

MODELS / PHYSICAL LIMITS

Competing geometries: what is excluded and what survives

TIC	Circular large min Delta chi2	Best exact eccentric e / Delta chi2	Diluted luminous min Delta chi2	Equal twins at 2P Delta chi2
193718989	85.2	0.836 / 2.7	-1.8	2.7
188830138	110.7	0.796 / 43.2	0.5	2.8
270573718	93.6	0.868 / -0.0	-11.5	-0.0
116046773	212.7	0.825 / 11.9	2.6	12.1
382098683	135.3	0.921 / 8.7	2.4	8.7
262843927	112.3	0.895 / 12.3	2.9	12.3
201247574	91.9	0.865 / 4.2	-2.7	4.3

Reference model: a small dark transiting body with freely fitted impact parameter and effective transit speed. "Large" means fixed radius 0.08, 0.10 or 0.12 solar radii. The circular cases fix the catalog target density. Their large likelihood penalties exclude these **specified circular, undiluted target-host models within the adopted likelihood**; they do not exclude every brown dwarf, low-mass star, or blend.

Exact Keplerian checks use apastron/periastron orientations, impose non-collision, and keep catalog semimajor axis. They demonstrate surviving geometries; they are not a complete eccentric posterior. Companion mass is neglected in catalog a/R, so a radius label cannot be promoted to a mass classification. TIC 188830138 retains a large penalty even at its eccentricity boundary; several longer-period candidates do not.

Nine luminous binary configurations per target couple primary and secondary eclipses at P, with radius ratios 0.2/0.5/1, surface-brightness ratios 0.01/0.1/1, free effective density, and physical third-light fractions between 0 and 1. Equal twins are fitted explicitly at 2P. A negative Delta chi-square can reflect a residual half-phase feature; it is not proof of a binary.

Odd/even agreement cannot reject equal twins. Conversely, the geometric grid does not impose stellar-evolution tracks and is not an exhaustive triple likelihood. The separate population calculations include bound/background/nearby planet and binary scenarios, including doubled periods. No high-resolution imaging, eclipse color, or RV mass information is invented for those likelihoods.

The radius profiles re-optimize impact parameter and speed. They provide finite sampled intervals for TIC 262843927, whose earlier curvature error was singular. Several other profiles reach the scan boundary; those endpoints are not upper confidence limits. All profile radii hold the catalog stellar radius fixed.

CALIBRATION / COMPLETENESS

Secondary searches with a measured null and recovery curve

TIC	Strongest phase	Nominal S/N	Global p	Same-width recovery depth 95% observed fraction (ppt)
193718989	0.8154	4.58	0.261	1.842
188830138	0.3496	3.81	0.048	0.200
270573718	0.9215	4.72	0.285	1.499
116046773	0.2148	5.61	0.050	0.500
382098683	0.8506	3.87	0.273	0.300
262843927	0.3870	8.89	0.493	2.000
201247574	0.4292	4.42	0.563	2.000

The primary is masked before detrending. Searches cover three widths (0.5, 1 and 2 times the primary duration) at fixed period. Five hundred independent sector-phase shifts calibrate the maximum across phase and width, preserving sector structure. This is a conditional search calibration, not the original survey detection false-alarm probability. The smallest resolved p-value is 1/501.

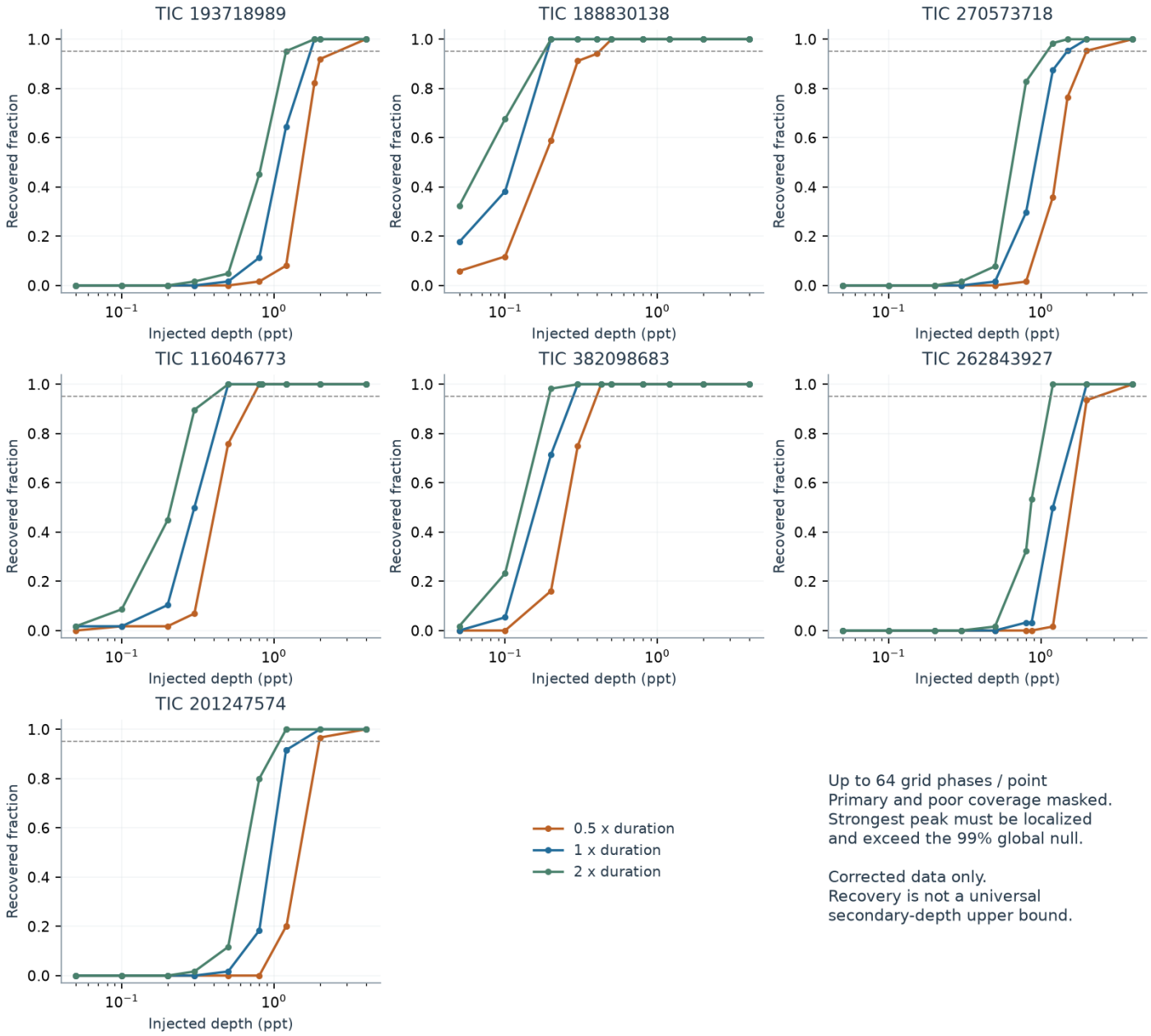
Signals are injected before detrending at up to 64 evenly spaced phases for each depth/width; primary-adjacent and poorly sampled phases are excluded, with actual denominators saved. Recovery requires the strongest peak to lie near the injected phase and exceed the 99% global null threshold. Wilson intervals are saved. The tabulated depth is an observed 95% recovery crossing, not a 95%-confidence completeness bound.

Corrected data support the most useful sensitivity. Saved and freshly extracted raw apertures can contain large residual systematics: several never reach 95% recovery even at the largest injected depth. Their weak limits are retained. A low corrected half-phase p-value for TIC 270573718 or TIC 201247574 does not survive as a stable secondary across reductions.

TIC	Half-phase depth (ppt, corrected)	Conditional p at phase 0.5	Half-phase upper (ppt, corrected)	Half-phase upper (ppt, fresh aperture)
193718989	0.308 +/- 0.201	0.086	0.709	0.679
188830138	-0.007 +/- 0.027	0.609	0.044	0.148
270573718	0.576 +/- 0.147	0.004	0.853	1.143
116046773	-0.004 +/- 0.046	0.547	0.095	0.194
382098683	-0.046 +/- 0.034	0.866	0.065	0.179
262843927	0.044 +/- 0.092	0.355	0.270	23.911
201247574	0.548 +/- 0.135	0.002	0.854	4.757

Half-phase values are fixed-phase diagnostics. Their one-sided upper estimates use the empirical null scale and cannot substitute for cross-reduction confirmation. The much weaker fresh-aperture bounds expose reduction sensitivity.

Injection recovery across all seven candidates



The ordinate is the fraction recovered among the admitted phase-grid locations, not the fraction of every possible phase. Exact denominators and Wilson intervals accompany each point in the JSON. Primary-adjacent phases and phases with nominal depth error above 5 ppt are omitted. The strongest recovered peak must be close to the injected phase as well as exceed the global threshold.

The crossings depend on duration: short eclipses are harder to recover. The curves include the actual residual data, so coherent artifacts can steal the strongest-peak location. This conservative behavior explains why a visually prominent feature need not be a robust secondary, and why raw-aperture recovery can remain poor.

POPULATION MODELS

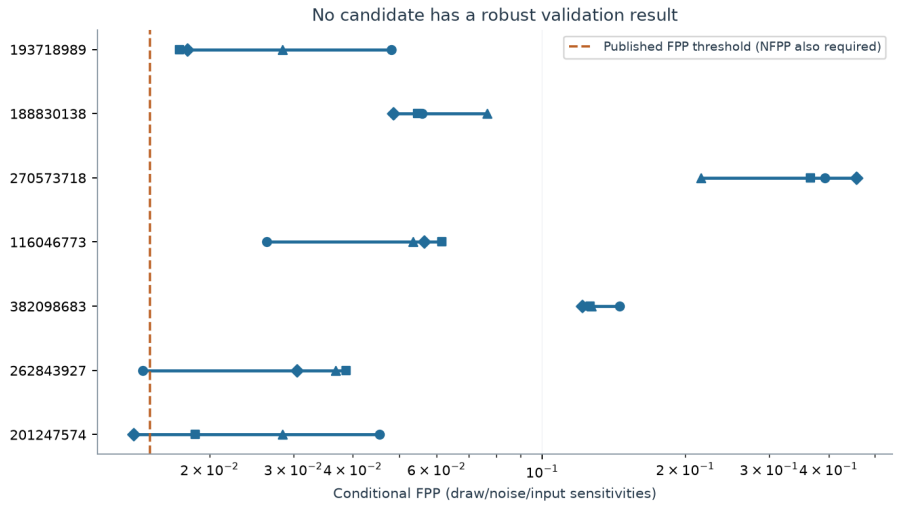
FPP sensitivity: conditional probabilities, not validation

TIC	FPP range	NFPP range	Relative-prior sensitivity FPP range
193718989	1.73% - 4.81%	0.000896 - 0.00475	0.17% - 42.29%
188830138	4.86% - 7.66%	0.00837 - 0.00991	0.45% - 63.94%
270573718	21.54% - 45.73%	0.187 - 0.438	1.13% - 98.62%
116046773	2.64% - 6.15%	9.67e-08 - 7.47e-05	0.29% - 39.59%
382098683	12.12% - 14.56%	5.21e-05 - 0.0013	1.37% - 63.06%
262843927	1.45% - 3.87%	0 - 0	0.15% - 28.68%
201247574	1.38% - 4.56%	0.00152 - 0.00631	0.13% - 46.66%

The installed, locally customized TRICERATOPS implementation is identified by file hash. Initial runs use 50,000 and 100,000 draws with different seeds, plus a 1.5x noise run. A 100,000-draw single native-cadence binning run tests extra smoothing from the initial event-bin/phase-bin workflow. Twenty-seven relative prior combinations rescale unresolved-companion planets, nearby scenarios and binaries by 0.1/1/10. Ranges are sensitivity envelopes, not credible intervals.

All scenario families are retained, with an actual Gaia-derived background. Gaia-undetected background stars remain incomplete. No measured contrast curve is available. The catalog-corrected light curve is assumed target-dilution-corrected; scene flux fractions are reused for six targets and recalculated with the mission PRF for TIC 270573718. Stellar catalog, aperture, integration and white-noise assumptions limit absolute calibration.

FPP = 1 - (TP + PTP + DTP) under the package convention. STP means a planet transiting an unresolved secondary star and therefore contributes to FPP; it is not an eclipsing binary. NFPP sums nearby-source cases. Several low FPP values are dominated by STP uncertainty or unstable Monte Carlo tails. TIC 262843927 has NFPP=0 because no neighbor passes the cached-throughput admission rule, not because observations prove zero nearby risk.



The commonly used validation thresholds are FPP < 0.015 and NFPP < 0.001 [1]. No candidate passes robustly across these tests and the independent observational caveats. Image scores and ZTF limits are not multiplied into FPP as if independent, fully calibrated likelihoods.

EXTERNAL EVIDENCE

Stellar properties, archival images, spectra, and activity

TIC	Gaia RUWE	SED R_star (solar)	Gaia RV variability p	Circular ellipsoidal mass scale (M_Jup)
193718989	1.04	0.526	0.183	23586.5
188830138	0.99	0.718	0.660	3.9
270573718	1.12	0.461	unavailable	61337.5
116046773	0.89	0.693	0.295	803.1
382098683	1.28	0.614	0.191	131.8
262843927	1.31	0.577	0.977	2178.7
201247574	1.16	0.617	0.943	25324.2

SED fits use Gaia G/BP/RP and 2MASS J/H/Ks, a published empirical dwarf sequence, 0.04-0.06 mag model floors, free normalization and extinction. A two-dwarf sequence is fitted as a diagnostic. These are consistency fits, not atmosphere/isochrone posteriors; metallicity, age, reddening and blending are not fully marginalized. AllWISE colors and quality flags are saved separately. Mean XP spectra are available for all seven; RVS products are available for three.

All targets have non-single-star flag zero in the queried Gaia release; the NSS table queries yield no matching solutions. Normal RUWE or an empty NSS query cannot exclude companions. The 9-arcsec companion to TIC 116046773 is consistent in parallax and proper motion, so its faintness is not a justification to discard it.

DSS1 and DSS2 plates were inspected for every target. Motion to 2016 is only a few arcseconds and bright-star wings limit background constraints. Two malformed DSS timestamps were parsed by their valid calendar date instead of substituting an equinox. Pan-STARRS i stacks show the northern scenes, but TIC 188830138 is partly masked and TIC 116046773 has a completely masked central two-arcsec-radius core. No calibrated sub-resolution contrast curve follows from these images.

LAMOST for TIC 188830138 and GALAH/APOGEE for TIC 382098683 provide the only retrieved external epoch RV measurements. Gaia summary amplitudes and errors are not velocity time series. ESO spectral cone queries returned no matches, an archive-specific result rather than a universal absence statement.

Orbital modulation uses reconstructed pre-flattening flux, sector quadratic trends, harmonics at 2P/P/P2, and half-day/two-day cluster errors. The listed ellipsoidal scale assumes alpha=1.5, circular orbit, edge-on inclination, catalog density and no dilution. Only TIC 188830138 gives a useful low scale (~3.9 M_Jup); this is not a general companion-mass upper bound. Rotation peaks are descriptive and not globally calibrated.

PLANET CANDIDATE; TARGET FAVORED; VALIDATION INCOMPLETE

TIC 193718989

Period	Retrospective late S/N	Host data	Conditional FPP
24.532886 d	4.91 corrected / 2.31 aperture	3 sectors	1.73% - 4.81%

Updated evidence

Resolved ZTF photometry disfavors a same-depth deep eclipse on the nearest neighbor. No new independent transit confirmation.

Excluded only within specified models

Circular, undiluted target-host companions of 0.08-0.12 solar radii are poor fits: minimum Delta chi-square 85.2. This conditional geometric exclusion does not determine mass. The best exact eccentric example has e=0.836 and Delta chi-square 2.7. The explicit equal-twin 2P model has Delta chi-square 2.7.

Host and resolved photometry

The 9.8-arcsec neighbor would need a 74% eclipse under the equal-throughput RP proxy. Resolved ZTF r limits are much smaller, but only three sampled nights and detection-only photometry prevent a completeness claim for deep eclipses.

The closest tested neighbor has loss difference 6.1 to 7.7 versus the target; positive favors the target. Its conditional simulation tail p ranges from 0.000999 to 0.015. These are template/noise tests, not host probabilities.

Secondary, stellar, and RV checks

Strongest searched secondary: phase 0.8154, global p=0.261; same-duration 95% observed recovery crossing 1.842 ppt. Odd/even differences are 0.85 sigma (corrected) and 0.69 sigma (aperture). Gaia RUWE=1.04; empirical SED radius=0.526 solar radii.

Gaia RV summary is not significantly variable. No usable epoch RV orbit was found in the queried surveys.

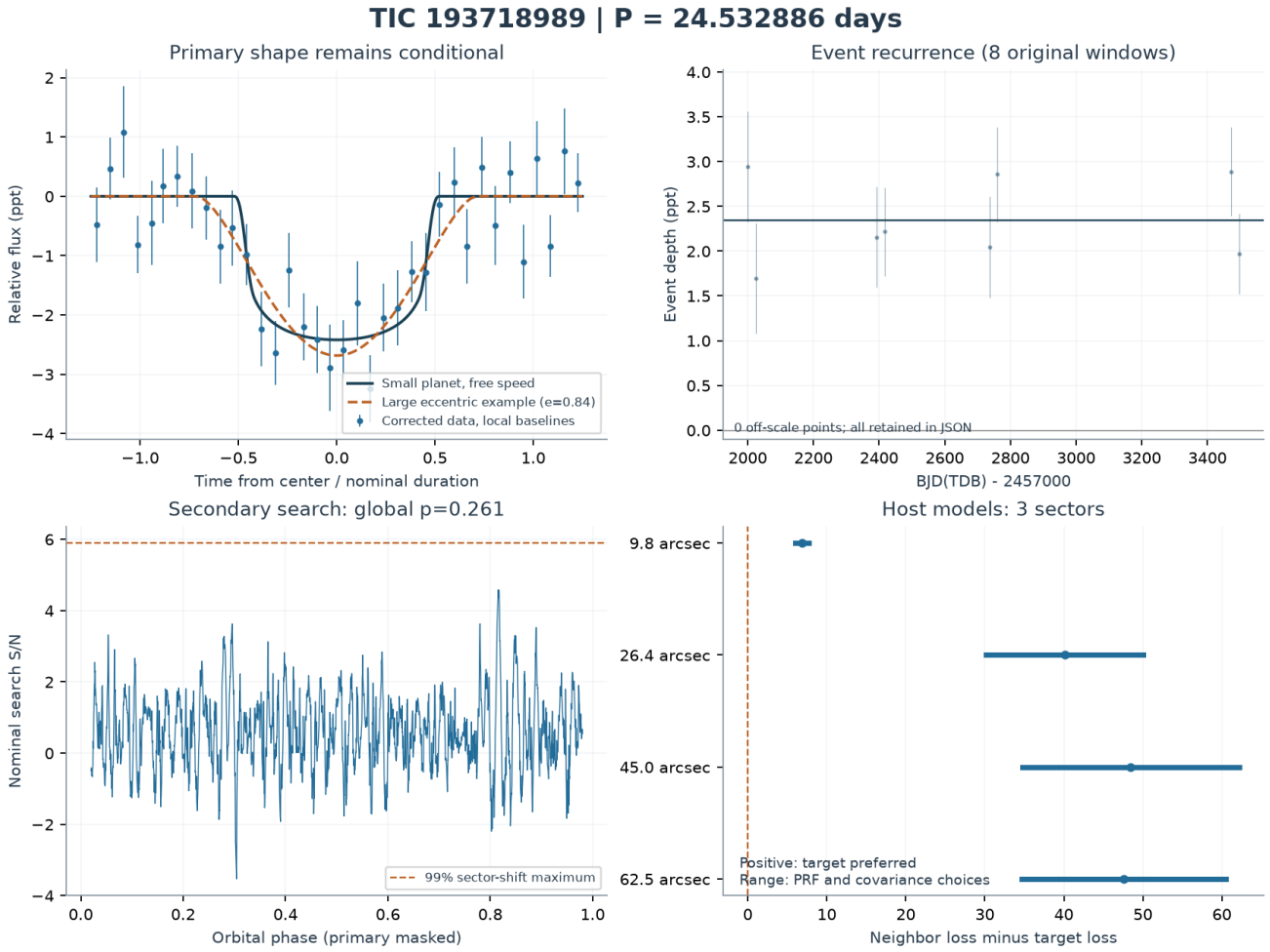
Still inconclusive / unavailable

Extreme eccentric grazing large companions, diluted binary/triple configurations, and unresolved companion-host planets remain. The profile radius interval reaches the sampled upper edge at Delta chi-square 4; it is not a finite upper confidence bound.

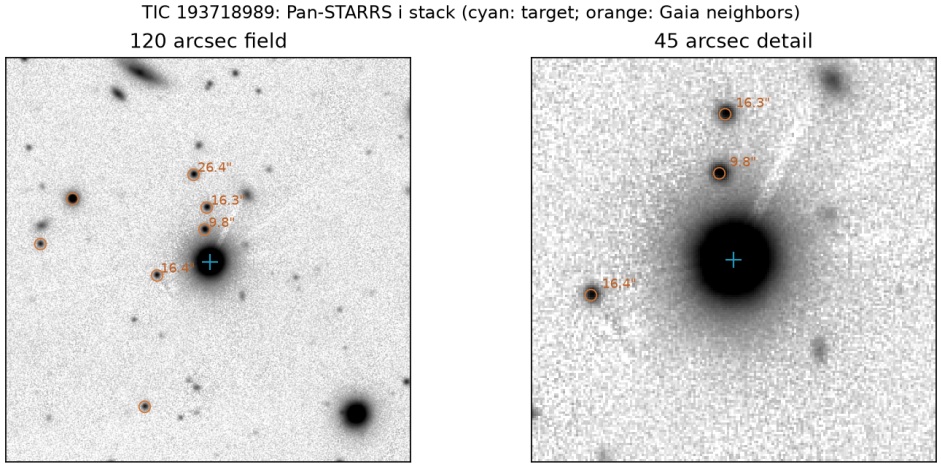
Decisive next observations: Resolved transit photometry and high-resolution imaging; additional RV epochs after a reconnaissance spectrum.

Evidence: analysis/193718989_*.json; exact models, localizations, calibrations, ground measurements, stellar checks and FPP scenario tables. Radius-profile Delta chi-square 4 grid interval: 2.43-4.79 Earth radii at fixed stellar radius; consult grid-edge limitations.

TIC 193718989: diagnostic evidence



Primary curves share fitted event-local baselines; the eccentric curve is an allowed geometric example, not a mass classification. Event-depth plot retains all measurements in the JSON even if extreme points fall outside the display. The secondary panel uses corrected data. Host ranges span PRF and covariance choices.



Archival scene: Pan-STARRS i stack; masked cores and bright-star wings prevent a contrast-curve interpretation. DSS images are also included in the bundle.

PLANET CANDIDATE; MUCH STRONGER HOST LOCALIZATION; RV FLAG

TIC 188830138

Period	Retrospective late S/N	Host data	Conditional FPP
1.181024 d	8.71 corrected / 8.37 aperture	20 sectors	4.86% - 7.66%

Updated evidence

Localization expands from four selected sectors to 20 usable scene fits. Northern ZTF neighbors have substantial resolved phase coverage.

Excluded only within specified models

Circular, undiluted target-host companions of 0.08-0.12 solar radii are poor fits: minimum Delta chi-square 110.7. This conditional geometric exclusion does not determine mass. The best exact eccentric example has e=0.796 and Delta chi-square 43.2. The explicit equal-twin 2P model has Delta chi-square 2.8.

Host and resolved photometry

For the 12.7-arcsec neighbor, r/i depth limits near 8% are below the 35% TESS proxy requirement. The 51.1-arcsec high-RUWE star has a g-band limit near 0.18%, below the 0.37% equal-throughput proxy; redder-band limits are less decisive at that proxy depth. These are same-depth, same-duration, fixed-ephemeris tests on detected epochs, not universal EB exclusions.

The closest tested neighbor has loss difference 47.8 to 116.3 versus the target; positive favors the target. Its conditional simulation tail p ranges from 0.000999 to 0.000999. These are template/noise tests, not host probabilities.

Secondary, stellar, and RV checks

Strongest searched secondary: phase 0.3496, global p=0.048; same-duration 95% observed recovery crossing 0.200 ppt. Odd/even differences are 2.02 sigma (corrected) and 1.65 sigma (aperture). Gaia RUWE=0.99; empirical SED radius=0.718 solar radii.

LAMOST gives -30.08 +/- 3.02 km/s (2017-06-14), versus Gaia -20.15 +/- 0.69 km/s: a 3.21-formal-sigma discrepancy. Survey offsets and a single low-resolution epoch prevent an orbital or binary claim; follow-up is warranted.

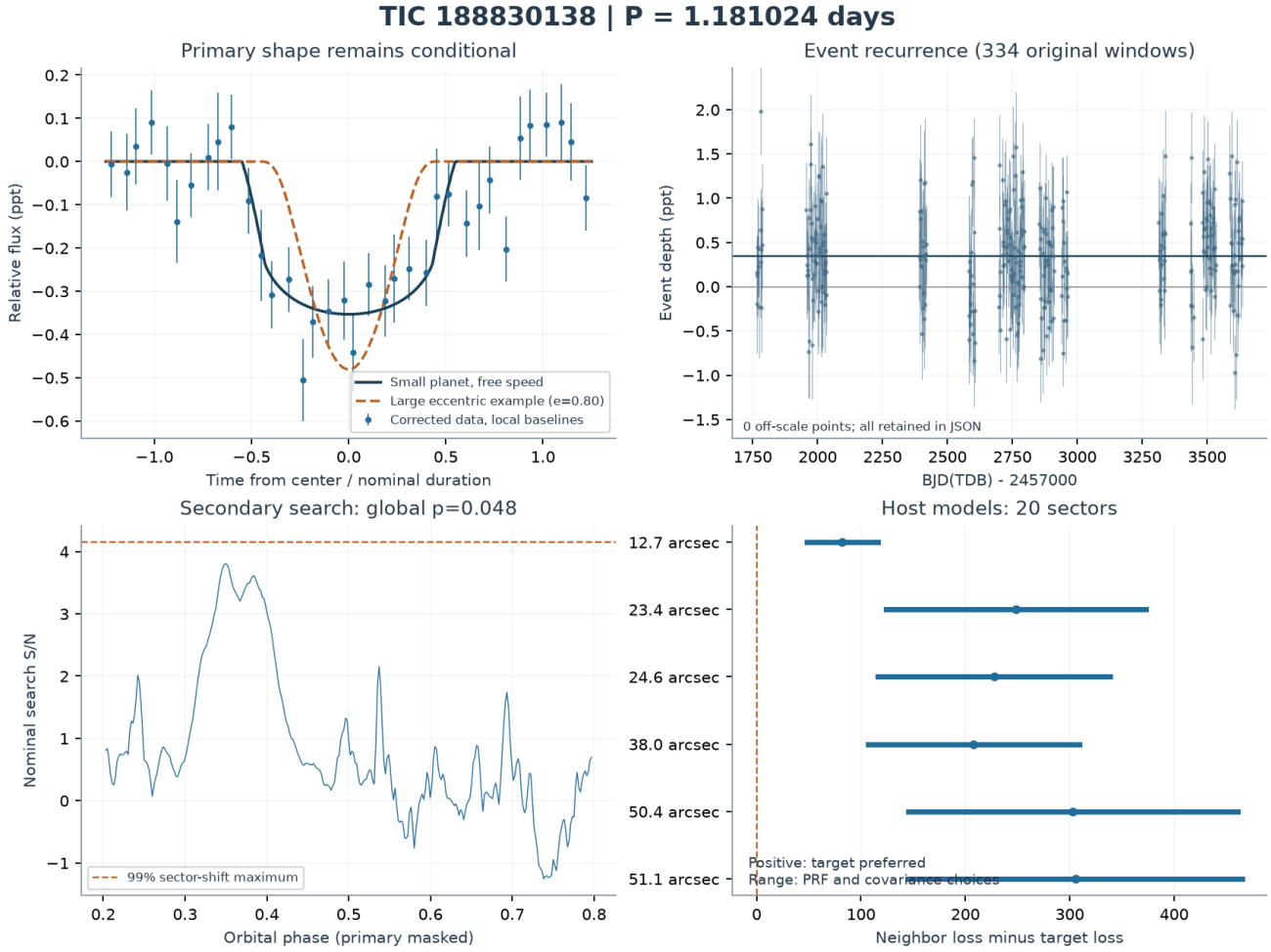
Still inconclusive / unavailable

An unresolved stellar companion hosting a planet still dominates the conditional FPP. Exact tested large dark companions remain poor fits even at the non-collision eccentricity boundary. The circular undiluted ellipsoidal mass scale is useful but does not apply to diluted/eccentric configurations.

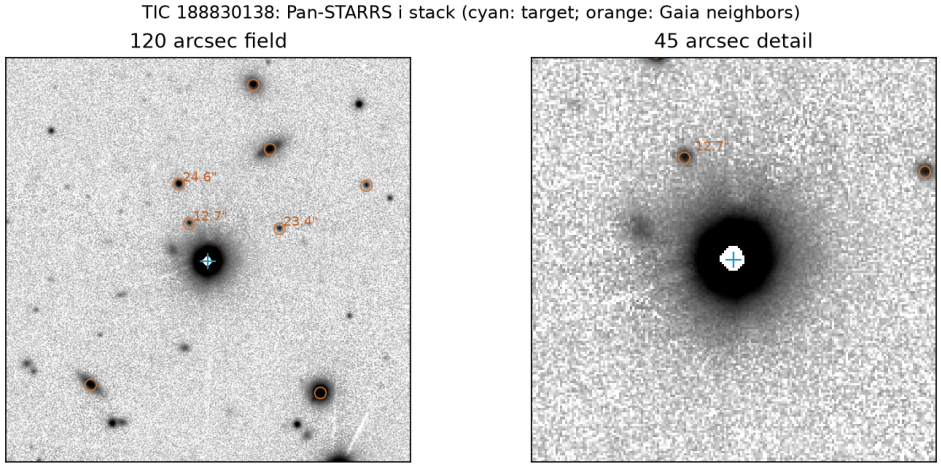
Decisive next observations: Repeat reconnaissance RVs to resolve the LAMOST discrepancy; high-resolution imaging and a resolved multiband transit.

Evidence: analysis/188830138_*.json; exact models, localizations, calibrations, ground measurements, stellar checks and FPP scenario tables. Radius-profile Delta chi-square 4 grid interval: 1.39-2.57 Earth radii at fixed stellar radius; consult grid-edge limitations.

TIC 188830138: diagnostic evidence



Primary curves share fitted event-local baselines; the eccentric curve is an allowed geometric example, not a mass classification. Event-depth plot retains all measurements in the JSON even if extreme points fall outside the display. The secondary panel uses corrected data. Host ranges span PRF and covariance choices.



Archival scene: Pan-STARRS i stack; masked cores and bright-star wings prevent a contrast-curve interpretation. DSS images are also included in the bundle.

UNVALIDATED CANDIDATE; MODEL AND TIMING AMBIGUITY INCREASED

TIC 270573718

Period	Retrospective late S/N	Host data	Conditional FPP
41.383701 d	0.29 corrected / 8.19 aperture	4 sectors	21.54% - 45.73%

Updated evidence

Previously missing complete FPP calculation now runs with a real Gaia background and fresh TIC/PRF scene. Its high nearby-source contribution conflicts with the target-favored image tests.

Excluded only within specified models

Circular, undiluted target-host companions of 0.08-0.12 solar radii are poor fits: minimum Delta chi-square 93.6. This conditional geometric exclusion does not determine mass. The best exact eccentric example has e=0.868 and Delta chi-square -0.0. The explicit equal-twin 2P model has Delta chi-square -0.0.

Host and resolved photometry

No resolved ground transit light curve was retrieved. Gaia epoch photometry has no nominal in-transit samples. The ExoFOP response was malformed, so the follow-up census remains incomplete.

The closest tested neighbor has loss difference 17.9 to 40.0 versus the target; positive favors the target. Its conditional simulation tail p ranges from 0.000999 to 0.000999. These are template/noise tests, not host probabilities.

Secondary, stellar, and RV checks

Strongest searched secondary: phase 0.9215, global p=0.285; same-duration 95% observed recovery crossing 1.499 ppt. Odd/even differences are 0.72 sigma (corrected) and 0.35 sigma (aperture). Gaia RUWE=1.12; empirical SED radius=0.461 solar radii.

Gaia supplies a mean RV, but usable variability diagnostics and a multi-epoch spectroscopic orbit were not retrieved.

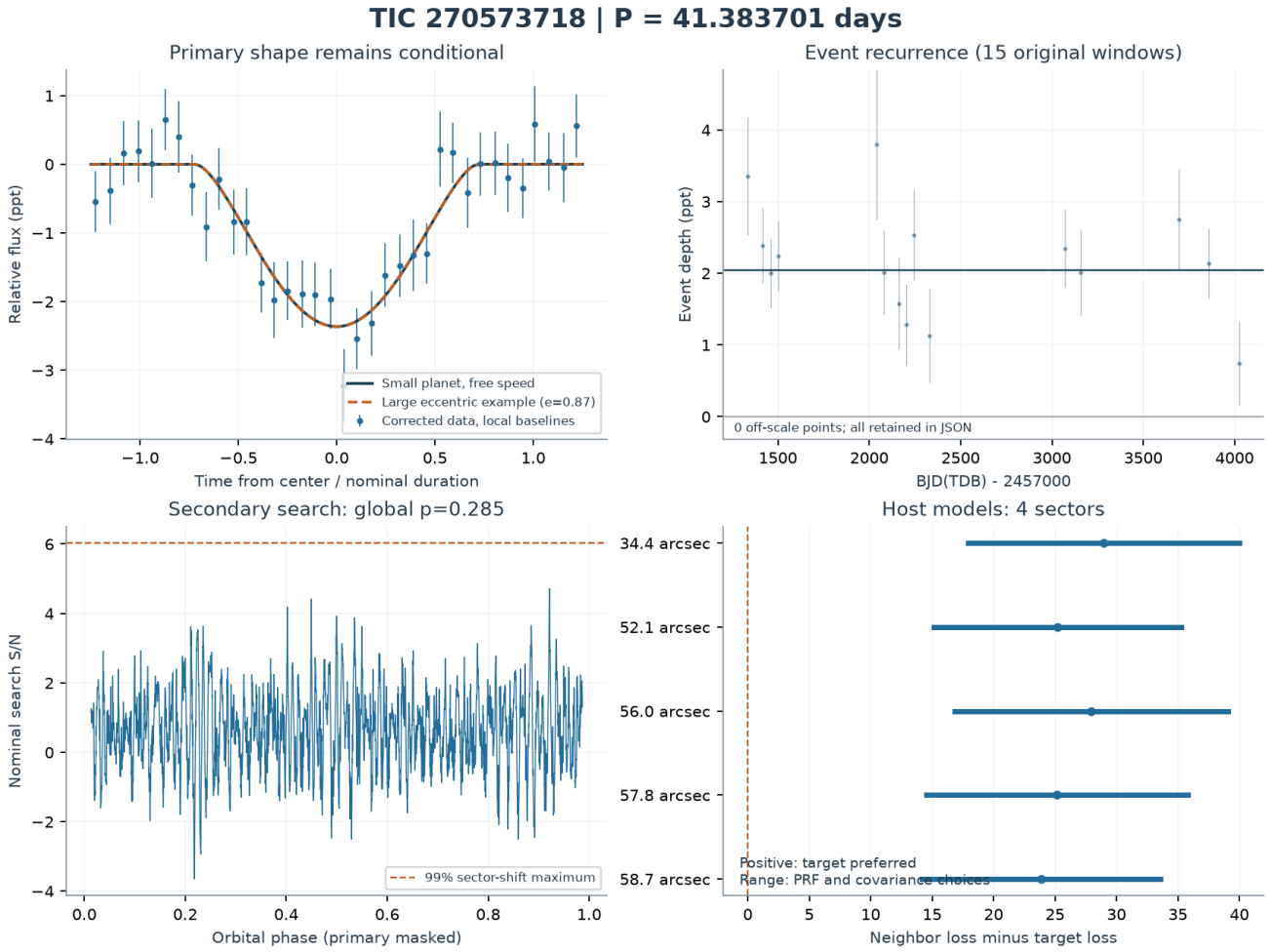
Still inconclusive / unavailable

Corrected-series early-to-late timing prediction fails while independent aperture extraction succeeds, indicating reduction/timing sensitivity. An exact e~0.865 grazing large-companion model fits almost as well as the free-speed small-planet geometry. A luminous diluted model improves the local half-phase fit, but the secondary is not robust across reductions. Nearby EB and doubled-period configurations remain probabilistically material.

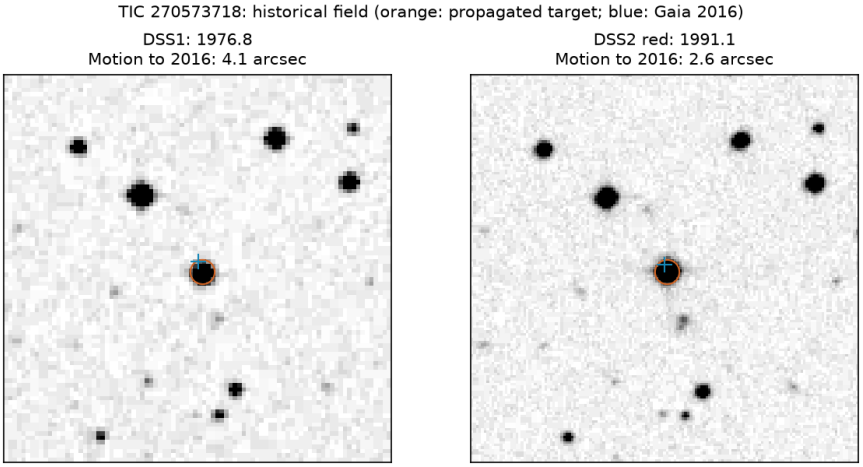
Decisive next observations: Prioritize resolved photometry at an updated long-period ephemeris and multiple RV epochs. Do not combine image Delta chi-square with FPP as independent likelihoods without a joint scene/noise model.

Evidence: analysis/270573718_*.json; exact models, localizations, calibrations, ground measurements, stellar checks and FPP scenario tables. Radius-profile Delta chi-square 4 grid interval: 2.07-3.83 Earth radii at fixed stellar radius; consult grid-edge limitations.

TIC 270573718: diagnostic evidence



Primary curves share fitted event-local baselines; the eccentric curve is an allowed geometric example, not a mass classification. Event-depth plot retains all measurements in the JSON even if extreme points fall outside the display. The secondary panel uses corrected data. Host ranges span PRF and covariance choices.



Archival scene: DSS1/DSS2; orange marks the propagated position, blue the Gaia-2016 position. Small displacement and plate PSFs limit background exclusion.

RECURRING SIGNAL; TARGET VERSUS BOUND COMPANION UNRESOLVED

TIC 116046773

Period	Retrospective late S/N	Host data	Conditional FPP
5.650272 d	12.30 corrected / 9.21 aperture	7 sectors	2.64% - 6.15%

Updated evidence

Seven-sector control-map experiments strengthen the diagnosis of PRF dependence, rather than resolving the 9-arcsec host ambiguity.

Excluded only within specified models

Circular, undiluted target-host companions of 0.08-0.12 solar radii are poor fits: minimum Delta chi-square 212.7. This conditional geometric exclusion does not determine mass. The best exact eccentric example has e=0.825 and Delta chi-square 11.9. The explicit equal-twin 2P model has Delta chi-square 12.1.

Host and resolved photometry

The close common-parallax companion needs roughly 9-12% eclipses. Resolved ZTF r reaches only about a 15% upper limit after three-sigma ephemeris-corner checks, so it cannot reject that scenario.

The closest tested neighbor has loss difference -55.7 to 25.1 versus the target; positive favors the target. Its conditional simulation tail p ranges from 0.000999 to 1. These are template/noise tests, not host probabilities.

Secondary, stellar, and RV checks

Strongest searched secondary: phase 0.2148, global p=0.050; same-duration 95% observed recovery crossing 0.500 ppt. Odd/even differences are 0.27 sigma (corrected) and 1.61 sigma (aperture). Gaia RUWE=0.89; empirical SED radius=0.693 solar radii.

Gaia does not show significant aggregate RV variability, but its published mean is not an orbit or brown-dwarf mass exclusion.

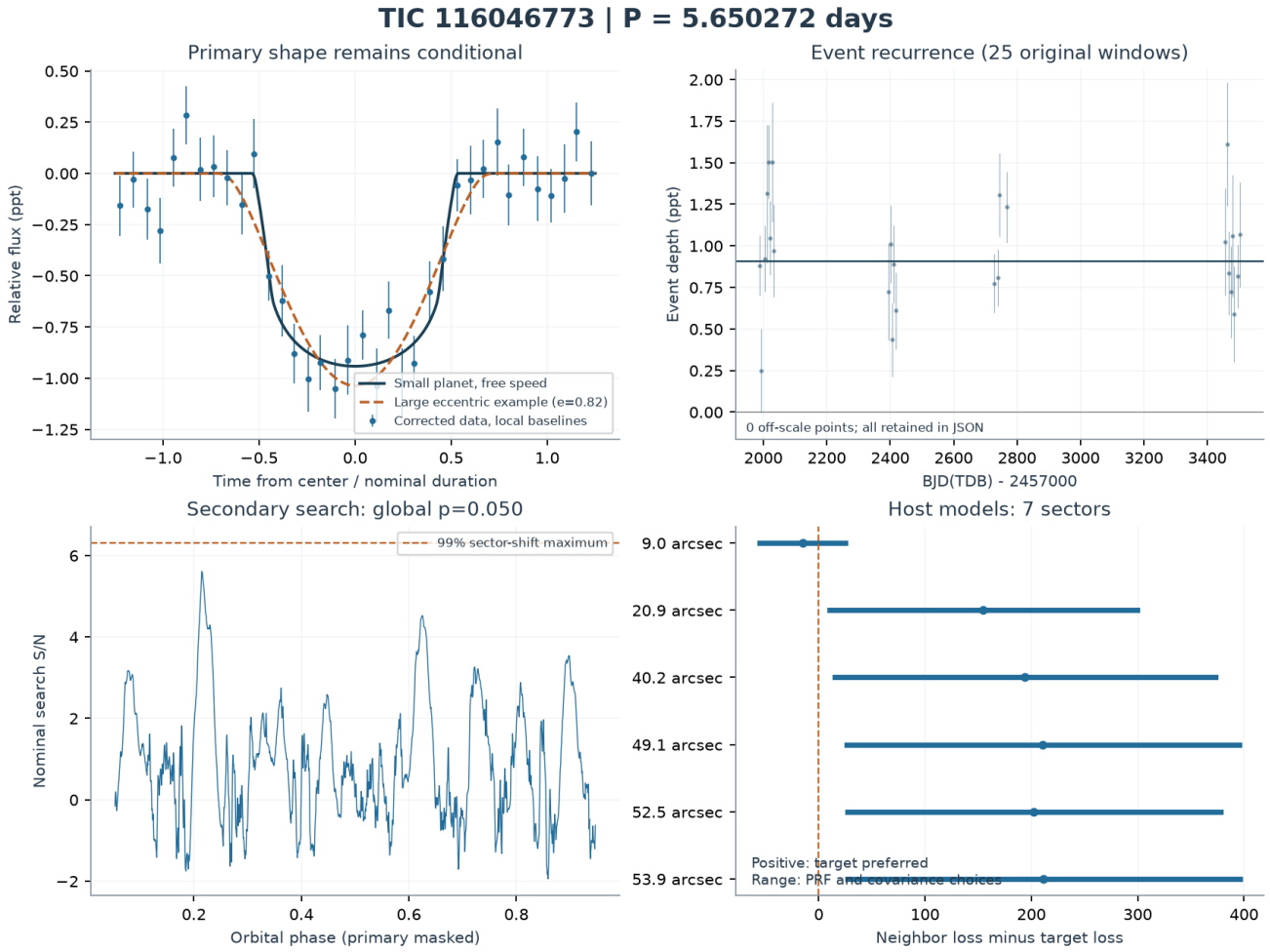
Still inconclusive / unavailable

Mission and empirical PRFs give opposite preferences between the target and 9-arcsec companion. Injected-host recovery depends strongly on the template family. A low conditional NFPP cannot override this direct scene ambiguity. The two-dwarf SED fit is comparably acceptable.

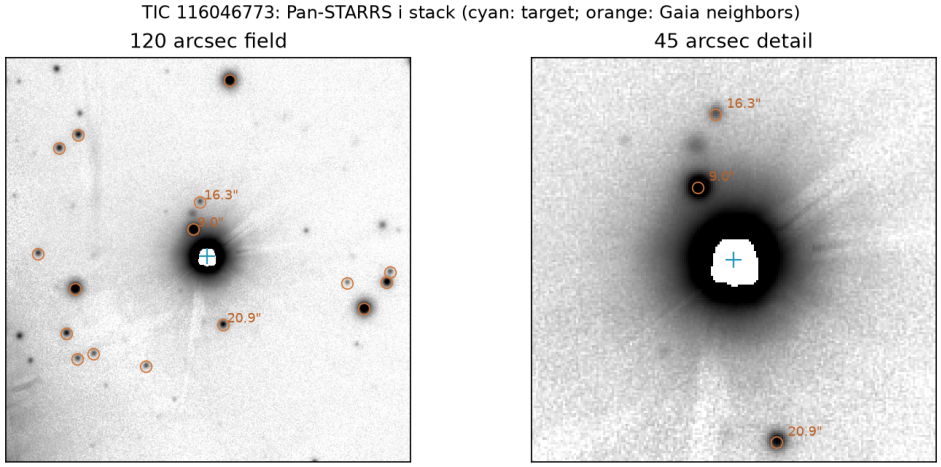
Decisive next observations: Obtain resolved eclipse photometry of both stars at 5.650272 days, with matched out-of-event coverage; obtain companion-aware spectra/RVs.

Evidence: analysis/116046773_*.json; exact models, localizations, calibrations, ground measurements, stellar checks and FPP scenario tables. Radius-profile Delta chi-square 4 grid interval: 2.09-2.91 Earth radii at fixed stellar radius; consult grid-edge limitations.

TIC 116046773: diagnostic evidence



Primary curves share fitted event-local baselines; the eccentric curve is an allowed geometric example, not a mass classification. Event-depth plot retains all measurements in the JSON even if extreme points fall outside the display. The secondary panel uses corrected data. Host ranges span PRF and covariance choices.



Archival scene: Pan-STARRS i stack; masked cores and bright-star wings prevent a contrast-curve interpretation. DSS images are also included in the bundle.

PLANET CANDIDATE; ROBUST RESOLVED-HOST LOCALIZATION; FPP REMAINS ELEVATED

TIC 382098683

Period	Retrospective late S/N	Host data	Conditional FPP
4.351685 d	7.09 corrected / 5.97 aperture	42 sectors	12.12% - 14.56%

Updated evidence

Localization expands from four selected sectors to 42. Archival GALAH/APOGEE measurements and a GALAH H-alpha-region spectrum are now incorporated.

Excluded only within specified models

Circular, undiluted target-host companions of 0.08-0.12 solar radii are poor fits: minimum Delta chi-square 135.3. This conditional geometric exclusion does not determine mass. The best exact eccentric example has e=0.921 and Delta chi-square 8.7. The explicit equal-twin 2P model has Delta chi-square 8.7.

Host and resolved photometry

No useful resolved ground transit series was retrieved. The low secondary-search sensitivity does not establish transit achromaticity.

The closest tested neighbor has loss difference 138.0 to 254.9 versus the target; positive favors the target. Its conditional simulation tail p ranges from 0.000999 to 0.000999. These are template/noise tests, not host probabilities.

Secondary, stellar, and RV checks

Strongest searched secondary: phase 0.8506, global p=0.273; same-duration 95% observed recovery crossing 0.300 ppt. Odd/even differences are 0.36 sigma (corrected) and 1.25 sigma (aperture). Gaia RUWE=1.28; empirical SED radius=0.614 solar radii.

Three GALAH RVs cover two dates. The constant-RV p-value is 0.035 with formal errors, 0.070 with a 0.1 km/s floor and 0.278 with a 0.3 km/s floor. APOGEE gives corrected RV -6.248 km/s; its raw -8.322 km/s is not directly comparable. These sparse data do not establish an orbit. Both surveys indicate metallicity near -0.30.

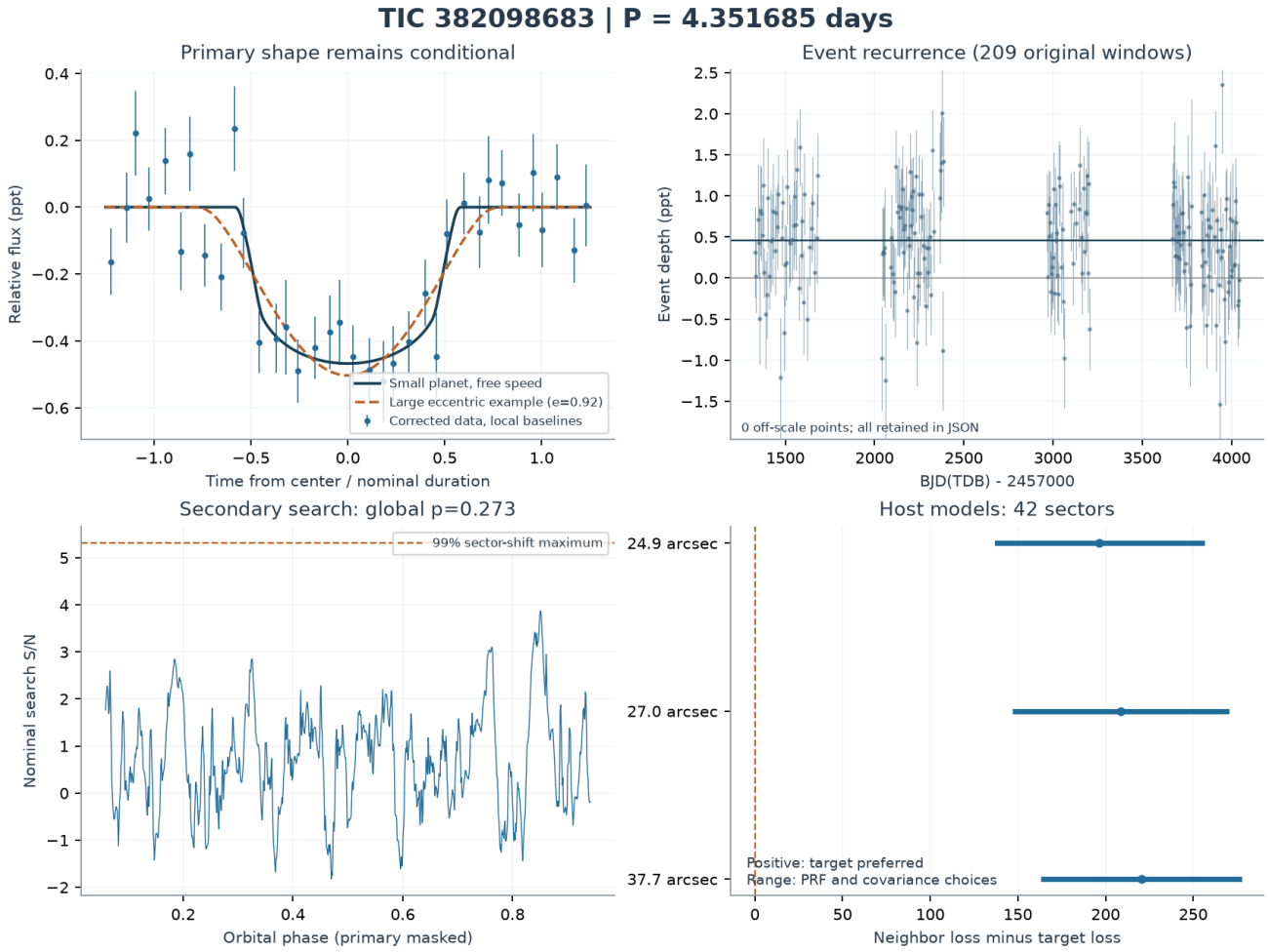
Still inconclusive / unavailable

The unresolved-companion planet scenario dominates the approximately 12-15% conditional FPP. Large eccentric grazing models are disfavored but not excluded as a class. Sparse spectra lack a calibrated secondary-line detection efficiency.

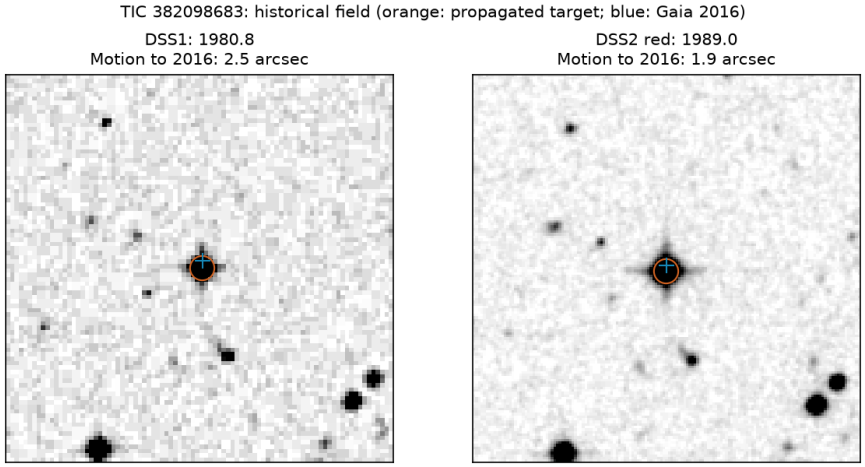
Decisive next observations: High-resolution imaging plus precise, phase-spread RVs; resolved multiband transit observations.

Evidence: analysis/382098683_*.json; exact models, localizations, calibrations, ground measurements, stellar checks and FPP scenario tables. Radius-profile Delta chi-square 4 grid interval: 1.27-1.81 Earth radii at fixed stellar radius; consult grid-edge limitations.

TIC 382098683: diagnostic evidence



Primary curves share fitted event-local baselines; the eccentric curve is an allowed geometric example, not a mass classification. Event-depth plot retains all measurements in the JSON even if extreme points fall outside the display. The secondary panel uses corrected data. Host ranges span PRF and covariance choices.



Archival scene: DSS1/DSS2; orange marks the propagated position, blue the Gaia-2016 position. Small displacement and plate PSFs limit background exclusion.

PLANET CANDIDATE; SECONDARY PEAK FAILS CALIBRATION

TIC 262843927

Period	Retrospective late S/N	Host data	Conditional FPP
24.108929 d	9.09 corrected / 4.11 aperture	3 sectors	1.45% - 3.87%

Updated evidence

The phase-0.387 feature has 8.9 nominal sigma but a sector-shift global p-value near 0.49. A profile-likelihood radius scan replaces the misleading zero curvature error.

Excluded only within specified models

Circular, undiluted target-host companions of 0.08-0.12 solar radii are poor fits: minimum Delta chi-square 112.3. This conditional geometric exclusion does not determine mass. The best exact eccentric example has e=0.895 and Delta chi-square 12.3. The explicit equal-twin 2P model has Delta chi-square 12.3.

Host and resolved photometry

No resolved ground transit light curve was retrieved. New archive sectors 31/88 are gross scene/photometric failures and provide no recurrence constraint.

The closest tested neighbor has loss difference 2.6 to 8.3 versus the target; positive favors the target. Its conditional simulation tail p ranges from 0.000999 to 0.000999. These are template/noise tests, not host probabilities.

Secondary, stellar, and RV checks

Strongest searched secondary: phase 0.3870, global p=0.493; same-duration 95% observed recovery crossing 2.000 ppt. Odd/even differences are 1.00 sigma (corrected) and 0.05 sigma (aperture). Gaia RUWE=1.31; empirical SED radius=0.577 solar radii.

Gaia RV diagnostics are consistent with no detected variability, with only a mean RVS spectrum available. No epoch RV orbit was retrieved.

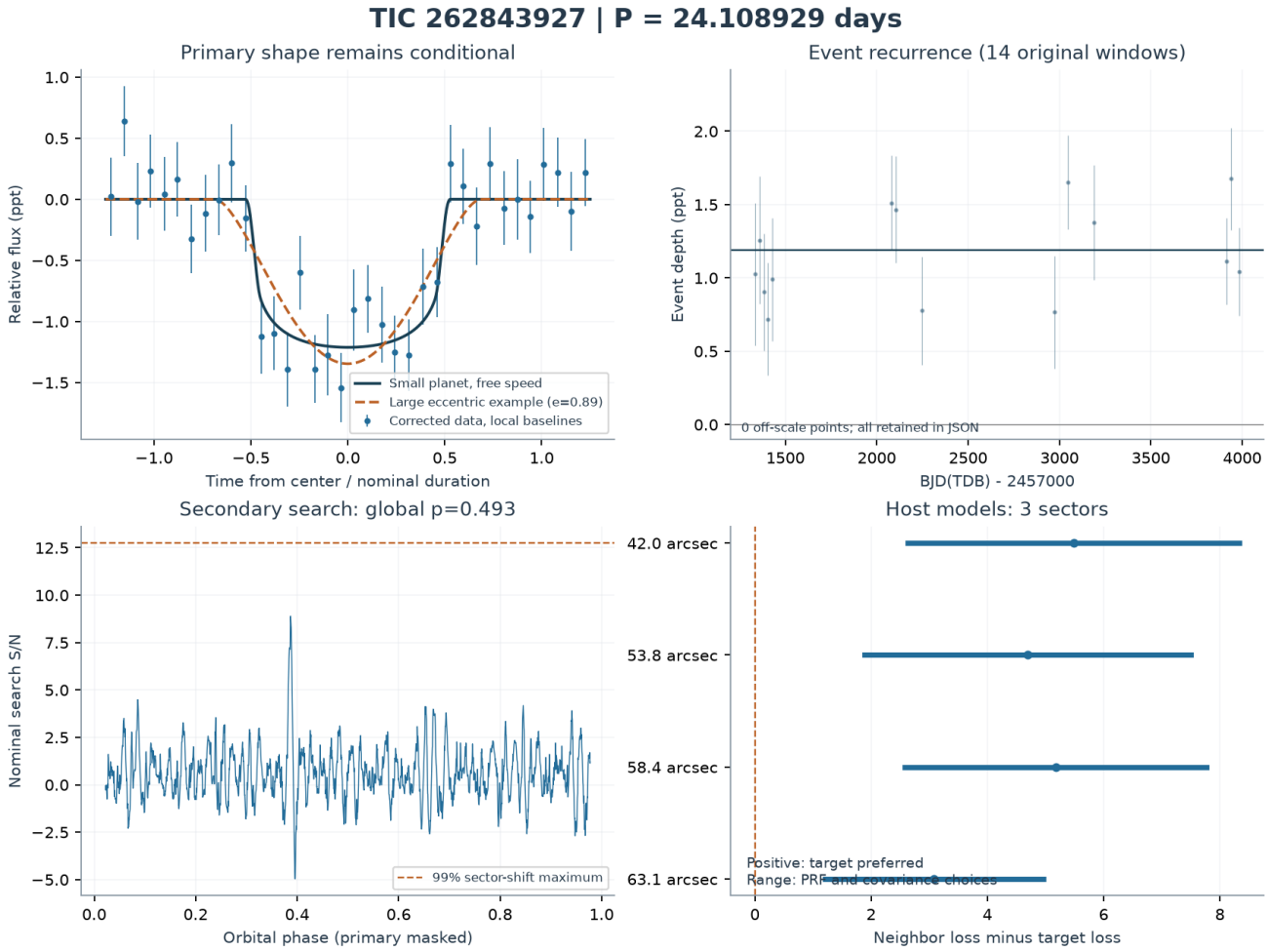
Still inconclusive / unavailable

NFPP=0 in the cached-throughput FPP scene is structural: no neighbor survives its flux-admission threshold. It is not an empirical proof of zero nearby-source risk. Monte Carlo runs straddle the FPP validation threshold. Eccentric/diluted models and unresolved companions remain.

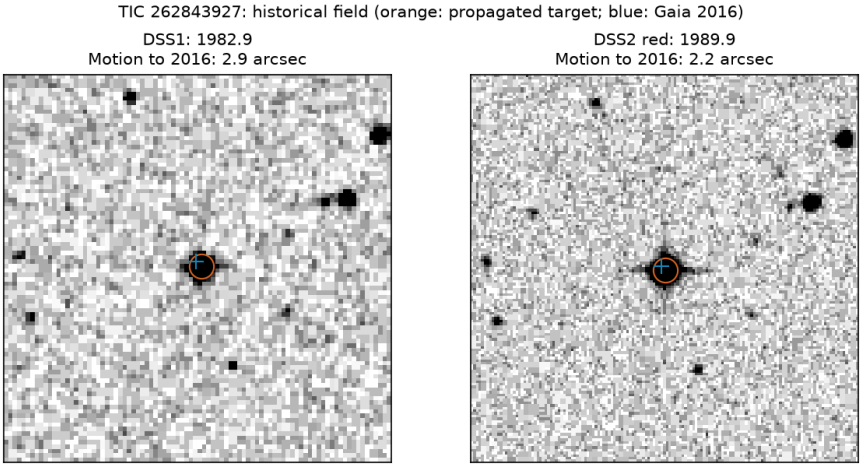
Decisive next observations: High-resolution imaging, resolved transit photometry and initial RV monitoring; retain finite radius-profile uncertainties.

Evidence: analysis/262843927_*.json; exact models, localizations, calibrations, ground measurements, stellar checks and FPP scenario tables. Radius-profile Delta chi-square 4 grid interval: 1.82-2.41 Earth radii at fixed stellar radius; consult grid-edge limitations.

TIC 262843927: diagnostic evidence



Primary curves share fitted event-local baselines; the eccentric curve is an allowed geometric example, not a mass classification. Event-depth plot retains all measurements in the JSON even if extreme points fall outside the display. The secondary panel uses corrected data. Host ranges span PRF and covariance choices.



Archival scene: DSS1/DSS2; orange marks the propagated position, blue the Gaia-2016 position. Small displacement and plate PSFs limit background exclusion.

PLANET CANDIDATE; LATER-DATA RECURRENCE INDEPENDENTLY STRENGTHENED

TIC 201247574

Period	Retrospective late S/N	Host data	Conditional FPP
16.560576 d	3.32 corrected / 3.08 aperture	5 sectors	1.38% - 4.56%

Updated evidence

Four previously unused events in TESS sectors 102 and 105 recur at the frozen ephemeris: pooled depth 1.56 +/- 0.30 ppt, S/N 5.13. This is new same-instrument evidence.

Excluded only within specified models

Circular, undiluted target-host companions of 0.08-0.12 solar radii are poor fits: minimum Delta chi-square 91.9. This conditional geometric exclusion does not determine mass. The best exact eccentric example has e=0.865 and Delta chi-square 4.2. The explicit equal-twin 2P model has Delta chi-square 4.3.

Host and resolved photometry

Gaia has only one nominal in-transit epoch in each of G/BP/RP, insufficient for an independent small-transit or color confirmation. No useful resolved ground series was retrieved.

The closest tested neighbor has loss difference 3.1 to 7.1 versus the target; positive favors the target. Its conditional simulation tail p ranges from 0.000999 to 0.00999. These are template/noise tests, not host probabilities.

Secondary, stellar, and RV checks

Strongest searched secondary: phase 0.4292, global p=0.563; same-duration 95% observed recovery crossing 2.000 ppt. Odd/even differences are 2.00 sigma (corrected) and 2.02 sigma (aperture). Gaia RUWE=1.16; empirical SED radius=0.617 solar radii.

Gaia summary RVs show no significant variability; no epoch spectroscopic orbit was retrieved. Long-timescale photometric activity weakens orbital-modulation limits.

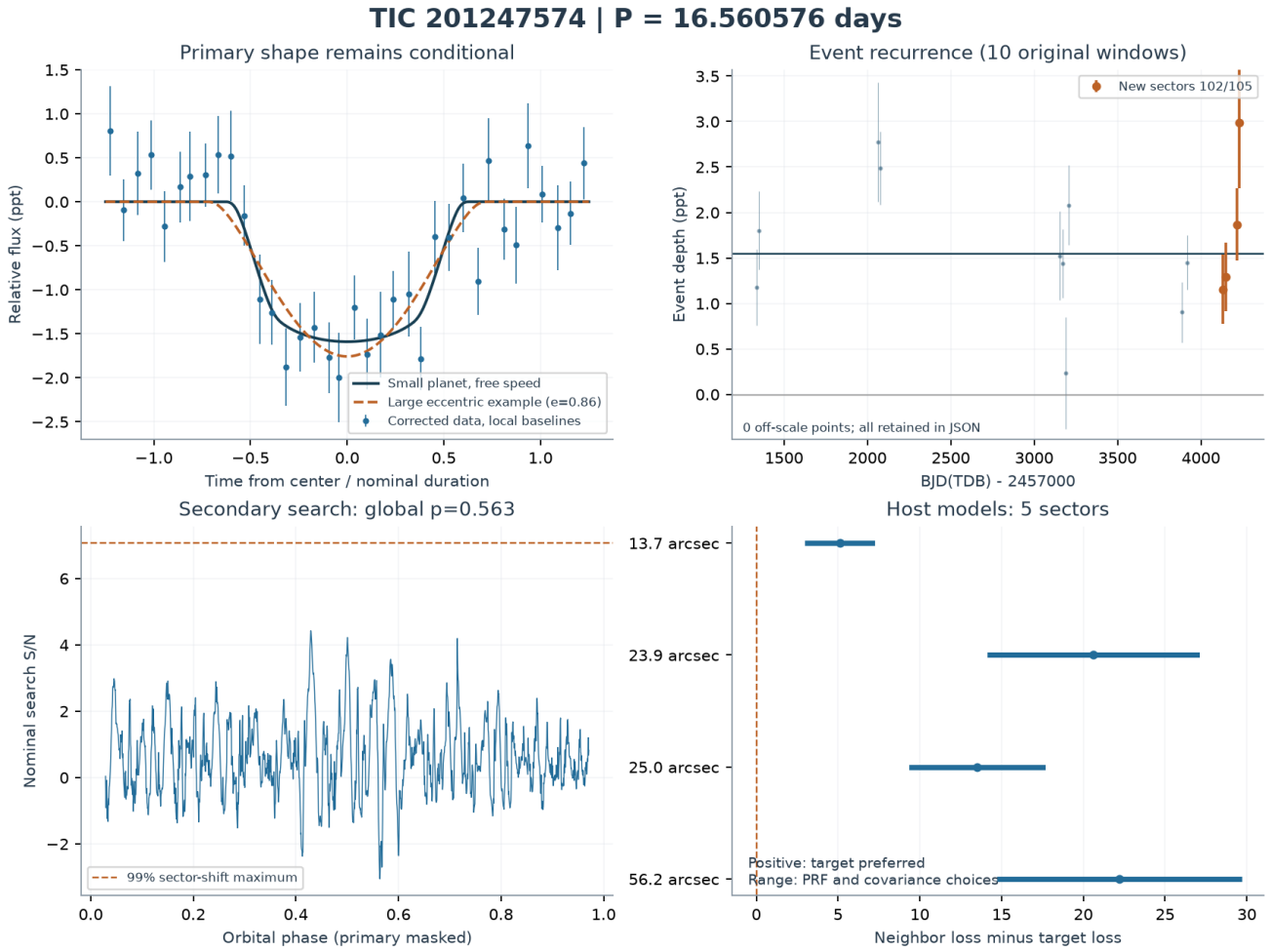
Still inconclusive / unavailable

The 13.7-arcsec neighbor remains less decisively excluded than farther neighbors. Exact high-eccentricity large companions and diluted binaries remain shape-degenerate. The corrected half-phase feature is not reproduced as a stable secondary in raw aperture reductions.

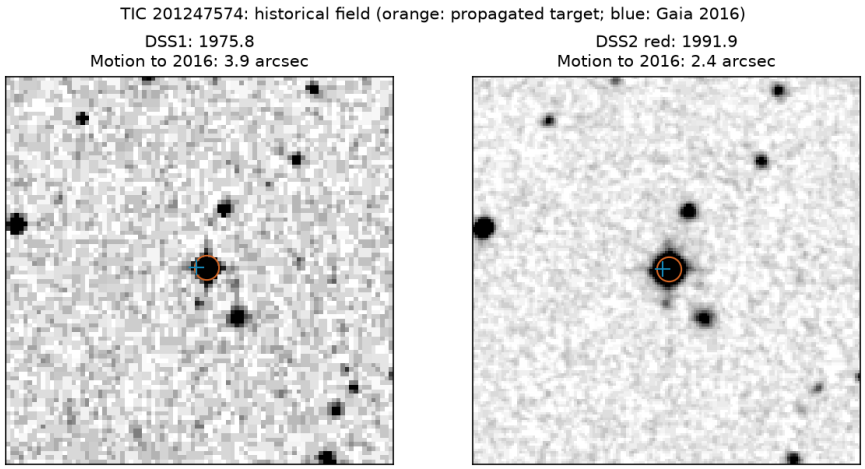
Decisive next observations: Resolved transit photometry of the target and 13.7-arcsec neighbor, high-resolution imaging and RV monitoring.

Evidence: analysis/201247574_*.json; exact models, localizations, calibrations, ground measurements, stellar checks and FPP scenario tables. Radius-profile Delta chi-square 4 grid interval: 2.23-4.39 Earth radii at fixed stellar radius; consult grid-edge limitations.

TIC 201247574: diagnostic evidence



Primary curves share fitted event-local baselines; the eccentric curve is an allowed geometric example, not a mass classification. Event-depth plot retains all measurements in the JSON even if extreme points fall outside the display. The secondary panel uses corrected data. Host ranges span PRF and covariance choices.



Archival scene: DSS1/DSS2; orange marks the propagated position, blue the Gaia-2016 position. Small displacement and plate PSFs limit background exclusion.

Resolved ground photometry: measured limits and their scope

TIC / Gaia source	Band	In-event nights minimum / nominal	Upper depth (percent)	TESS proxy (percent)
193718989 1346023547768414976	zg	2 / 2	39.83	73.72
193718989 1346023547768414976	zi	0 / 1	396.06	73.72
193718989 1346023547768414976	zr	3 / 3	20.29	73.72
188830138 1416467677212567424	zg	90 / 90	0.18	0.37
188830138 1416467677212567424	zi	26 / 26	0.47	0.37
188830138 1416467677212567424	zr	82 / 93	0.40	0.37
188830138 1416479423947847296	zg	3 / 4	44.08	34.92
188830138 1416479423947847296	zi	24 / 24	7.91	34.92
188830138 1416479423947847296	zr	47 / 48	7.69	34.92
116046773 4608570247290179328	zg	13 / 16	21.48	8.91
116046773 4608570247290179328	zi	3 / 4	20.16	8.91
116046773 4608570247290179328	zr	14 / 16	15.30	8.91

The bound is the maximum of conditional one-sided 95% depth limits at the nominal ephemeris and four +/-3-sigma epoch/period corners. No timing covariance was supplied, so this envelope is a sensitivity check rather than a joint confidence region. Duplicate exposures are removed, catflags=0 and magerr<0.3 are required, and semester/object baselines plus empirical jitter and night-cluster errors are used. MJD(UTC) is converted to BJD(TDB) at Palomar before folding.

The TESS proxy uses RP flux ratios and equal aperture throughput. Actual TESS dilution can demand deeper contaminant eclipses, but g/r/i color contrasts differ. These limits reject some **same-depth, same-duration models on the observed detection epochs**. They do not establish achromatic shallow transits or complete deep-eclipse recovery: ZTF non-detections are absent and no forced-photometry selection model is available.

Target-star ZTF light curves for TIC 193718989 and TIC 188830138 do not reach their shallow TESS transit depths. TIC 116046773 has no usable target table in this retrieval; its close companion is measured, but the r limit remains too weak. Gaia epoch photometry contributes zero nominal primary samples for TIC 193718989 and TIC 270573718, and one each in G/BP/RP for TIC 201247574.

What remains blocked, rather than "passed"

Test family	Completed here	Remaining limit

Recurrence

Completed: Time splits for all seven; four new later events for TIC 201247574.
Inconclusive or unavailable: Other candidates lack new independent observing confirmation; poor/empty sectors are not non-detections.

Localization

Completed: Correlated control maps, model cross-injections, offset perturbations and jackknife diagnostics.
Inconclusive or unavailable: Unresolved companions and missing catalog sources are outside the simulation; TIC 116046773 host remains ambiguous.

Ground / color

Completed: Resolved ZTF g/r/i fits for accessible northern targets and neighbors; Gaia epoch phase coverage.
Inconclusive or unavailable: No complete non-detection/forced-photometry model, no shallow multiband transit confirmation, and no useful southern resolved transit series.

Competing models

Completed: Geometric planet, large dark companion, luminous diluted binary, twins at 2P and exact eccentric examples; population alternatives.
Inconclusive or unavailable: No measured mass; no exhaustive stellar-evolution triple evidence integral or fully joint multiband/pixel/RV inference.

Secondary / recovery

Completed: Fixed-period global phase/width null and injections through detrending in three reductions.
Inconclusive or unavailable: Raw systematics weaken some limits; neither a universal secondary exclusion nor the discovery false-alarm probability.

Stellar / imaging

Completed: Six-band empirical SED, Gaia/NSS/neighbor astrometry, XP/RVS products, DSS and northern PS1.
Inconclusive or unavailable: No calibrated high-resolution contrast curve; empirical SEDs do not exclude faint companions or fully determine stellar ages/metallicity.

Spectra / activity

Completed: LAMOST, GALAH/APOGEE checks; Gaia RV summaries; orbital harmonics and variability.
Inconclusive or unavailable: No well-sampled precision RV orbit or secondary-line completeness test. Long-period ellipsoidal limits are weak.

FPP

Completed: All model families, real background inputs, seed/draw/noise/binning tests and relative-prior sensitivity.
Inconclusive or unavailable: No robust validation; background incompleteness, catalog errors and finite Monte Carlo tails persist.

The accompanying test_matrix.csv contains 77 candidate-by-test records with status, interpretation and evidence filenames. "Excluded" always names the specified model assumptions; no broad planet/binary/brown-dwarf/triple class is declared impossible from these data alone.

Reproducing and auditing the continuation

The evidence bundle includes the analysis scripts, derived JSON/CSV/NPZ tables, request receipts, public catalog/spectral/image responses, figures, version inventory, validation checks and manifest. Large TESS pixel cubes are retained locally and represented in the compact bundle by exact source paths, request records and hashes. The earlier corrected/aperture arrays and packet products remain dependencies at their recorded local paths; the bundle is not falsely presented as a standalone raw-data archive.

Stage	Entry points / outputs
Shared inputs	continuation_common.py; input_audit.json; environment.json; provenance manifest
Public data	fetch_archive_extension.py; fetch_archive_followups.py; fetch_spectra_and_tic.py; fetch_last_archives.py; fetch_spectra_detail.py; fetch_ps1_images.py
Pixels	fetch_additional_pixels.py; extract_additional_pixels.py; extract_fresh_apertures.py; localization_controls.py
Photometric models	fit_recurrence_models.py; exact_eccentric_and_timing.py; binary_joint_models.py; calibrated_searches.py
External tests	activity_and_ground.py; stellar_archival.py; followup_checks.py; inspect_ps1.py
FPP	fpp_sensitivity.py; fpp_native_bins.py; analysis/fpp/*_probabilities.csv and stellar scene tables
Synthesis / QA	assemble_evidence.py; make_figures.py; validate_evidence.py; build_continuation_report.py; reproduce.ps1

Run reproduce.ps1 with the astronomy Python path set to the installed tesshunter environment. The PDF uses the separate Codex artifact Python with ReportLab. Network fetching is opt-in in the reproduction script because it may be slow and archive behavior can change; cached responses preserve exactly what was analyzed. The locally customized TRICERATOPS files and empirical sequence are hashed and copied into provenance for inspection.

Scientific checks

The validation script checks all seven candidates and all requested test families, probability normalization, simulation denominators and p-value floors, finite model geometry, flux fractions, exact-orbit non-collision, time conversions, request/file hashes and a synthetic injected-decrement sign/amplitude test. It also audits the native-binning FPP run separately. PDF text and rendered pages were reviewed for truncation, clipping, blank pages and unreadable figures.

This work is an evidence audit and a reproducible set of follow-up analyses. Mass measurements, validated planetary radii on an identified host, and complete population probabilities require additional observations or a stronger joint inference model. No absent archive result, normal Gaia statistic, or low conditional FPP is substituted for that evidence.

PRIMARY SOURCES / ACCESS DETAILS

Methods and archive references

[1] **Giacalone et al., TRICERATOPS scenario framework and validation criteria**
https://arxiv.org/abs/2002.00691

[2] **NASA/HEASARC TESS data products and cadence documentation**
https://heasarc.gsfc.nasa.gov/docs/tess/data-products.html

[3] **MAST TESSCut public service**
https://mast.stsci.edu/tesscut/

[4] **IRSA ZTF light-curve API**
https://irsa.ipac.caltech.edu/docs/program_interface/ztf_lightcurve_api.html

[5] **Gaia DR3 epoch-photometry data model: BJD in TCB minus 2455197.5**
https://gea.esac.esa.int/archive/documentation/GDR3/Gaia_archive/chap_datamodel/sec_dm_photometry/ssec_dm_epoch_photometry.html

[6] **Gaia DR3 radial-velocity properties and validation**
https://arxiv.org/abs/2206.05902

[7] **Gaia DR3 astrophysical-parameter inference and limitations**
https://doi.org/10.1051/0004-6361/202243688

[8] **Pecaut & Mamajek empirical dwarf sequence, maintained table**
https://www.pas.rochester.edu/~emamajek/EEM_dwarf_UBVIJHK_colors_Teff.txt

[9] **Dawson & Johnson, photoeccentric interpretation of transit duration**
https://arxiv.org/abs/1203.5537

[10] **GALAH DR3 catalogues and best-method radial velocities**
https://www.galah-survey.org/dr3/the_catalogues/

[11] **GALAH DR3 spectra and data access**
https://www.galah-survey.org/dr3/the_spectra/

[12] **SDSS APOGEE: relative versus barycenter-corrected visit RVs**
https://www.sdss4.org/dr16/irspec/use-radial-velocities/

[13] **LAMOST DR5 release documentation**
https://dr5.lamost.org/v1/doc/release-note

[14] **ESO programmatic archive services**
https://archive.eso.org/programmatic/

[15] **Pan-STARRS image cutouts, linearized flux and timing caveats**
https://outerspace.stsci.edu/spaces/PANSTARRS/pages/298812251/PS1+Image+Cutout+Service

[16] **VizieR catalogs queried: II/246, II/328, J/MNRAS/506/150, III/286, V/164**
https://vizier.cds.unistra.fr/

[17] **DSS plate retrieval service**
https://archive.stsci.edu/cgi-bin/dss_form

Exact archive query URLs, filters, cone radii, source identifiers, UTC retrieval timestamps, raw responses and SHA-256 digests are retained in analysis/archives/*.request.json. A successful HTTP status is not itself treated as successful scientific data: malformed ExoFOP bodies and empty GALAH product VOTables are identified explicitly.